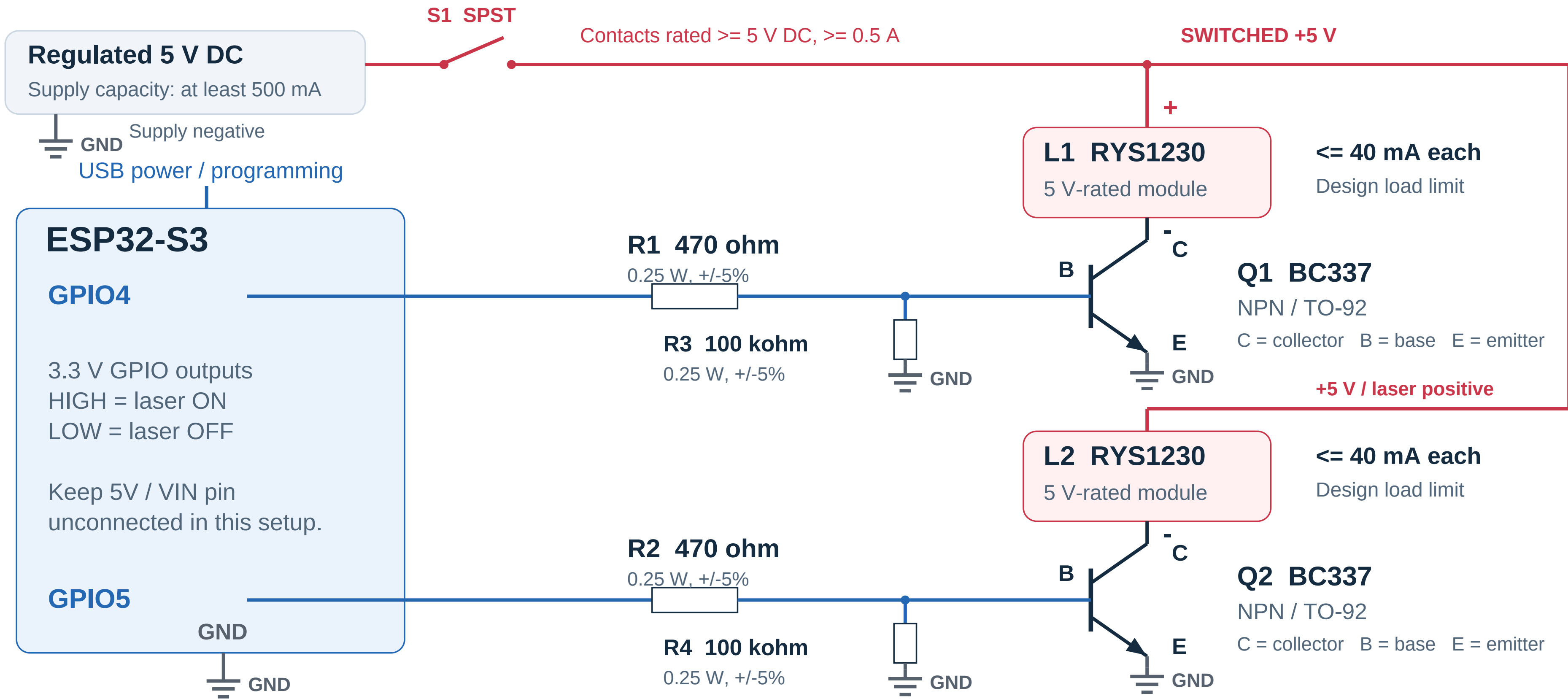


ESP32-S3 driving two RYS1230 lasers

Q1 / Q2 are BC337 NPN transistors. R1 / R2 must be 470 ohm when replacing the earlier MOSFET circuit.



All GND symbols connect together: ESP32 GND, supply negative, both emitters and both pull-down resistors.

Component ratings and wiring notes

Use one transistor per laser. Populate only Q1 / R1 / R3 / L1 if you have a single laser module.

Reference	Qty	Component / rating	Use in this circuit
Q1, Q2	2	BC337 NPN; 45 V / 800 mA / 625 mW max*	Actual design load <= 40 mA per transistor
R1, R2	2	470 ohm; 0.25 W (1/4 W); +/-5% or better	GPIO-to-base current limiting
R3, R4	2	100 kohm; 0.25 W (1/4 W); +/-5% or better	Base-to-emitter pull-down
L1, L2	2	RYS1230 complete module; 5 V input rating	Internal current limiting; verify your variant
Supply	1	Regulated 5 V DC; >= 500 mA capacity	Two laser branches: <= 80 mA combined
S1	1	SPST on/off; contacts >= 5 V DC and >= 0.5 A	Disconnects positive supply to both lasers

Connections and checks

Laser + -> switched 5 V. Laser - -> collector (C).
Emitter (E) -> GND. GPIO -> 470 ohm -> base (B).
100 kohm connects base to emitter / common GND.
Referenced onsemi part: pin 1 = C, pin 2 = B, pin 3 = E.
Confirm pin orientation against your exact manufacturer.
Use USB for ESP32 power; leave its 5V / VIN pin open.

Design limits

At 3.3 V GPIO and about 0.8 V base-emitter voltage:
base current ~= (3.3 - 0.8) / 470 = 5.3 mA per channel.
Base resistor dissipation ~= 0.013 W; 0.25 W is ample.
*VCEO / IC / PD maxima at 25 C; PD derates with temperature.
Check actual laser current <= 40 mA and proper turn-off.
Keep the full laser beam contained on a matte model surface.

Sources: [onsemi BC337 datasheet](#) [RYS1230 product specification](#) [ESP32-S3 board reference](#)

RYS1230 variants differ. This circuit assumes a complete 5 V-compatible module; it is not a bare laser-diode driver.

Model warning lights only | GPIO numbers are not physical header positions | Design checked; hardware not bench-tested